

Triplet optimization

Try to follow these steps:

1. Work your way through Section 9 of "tutorial_bmad_tao.pdf" found in the Bmad documentation. Goal: understand the difference between design/model/base lattices.
2. Work your way through Section 10 of the same tutorial. Goal: understand the basics of optimization in Bmad.
3. Finally, repeat the steps from Section 10 for our triplet.bmad model. The files are supplied with this module. Notice that the data block of the tao.init file is different now:

```
&tao_d2_data
  d2_data%name = 'elements'
  n_d1_data = 1
/

&tao_d1_data
  ix_d1_data = 1
  d1_data%name = 'r'
  datum(1) = 'r.12'      '' '' 'END'    'target' 0.0  1.0
  datum(2) = 'r.34'      '' '' 'END'    'target' 0.0  1.0
/
```

Make sure you understand what these changes imply, what "r.12" and "r.34" are, and why the aim is to get them both to zero.

4. Try the following commands with your triplet.bmad:

```
tao
show lattice
opt
show data
show data elements
show data elements[1]
show lattice
place r13 beta
place r23 beta
place r33 beta
set plot r13 component=design
```

```
set plot r23 component=model  
set plot r33 component=model-design  
scale
```

5. Think about the connection between this optimization routine and the imaging systems we discussed earlier. Bring lots of questions to our next class or share them by email.